

Supplementary Table S1. Between-segment distribution summary. Each continuous cell reports P5 / median / P95 across repeated action segments or occlusion episodes. Metrics whose names contain P95 are first computed within each segment over render frames. For $n = 4$, P5 and P95 are interpolated near the observed endpoints and should be interpreted descriptively.

Metric	Unit	n	Arrival-Hold	Capture-Hold	One-Euro Anchor	EgoAnchor
Head-motion leakage P95	mm	4	17.438 / 20.690 / 29.433	8.561 / 10.833 / 20.804	7.637 / 8.329 / 10.329	1.267 / 1.631 / 4.184
Absolute registration P95	mm	4	15.085 / 21.341 / 27.524	12.300 / 13.257 / 18.370	11.950 / 12.407 / 14.311	6.773 / 6.894 / 8.663
Stationary frame-increment P95	mm	4	2.374 / 3.760 / 5.738	1.269 / 2.380 / 3.479	0.655 / 1.065 / 1.771	0.081 / 0.098 / 0.112
Translation effective lag	ms	30	165.0 / 182.5 / 207.8	242.2 / 250.0 / 257.8	375.0 / 385.0 / 392.8	305.0 / 320.0 / 341.0
Translation lag-aligned RMSE	mm	30	7.539 / 10.087 / 11.415	7.304 / 9.584 / 11.257	7.165 / 9.559 / 11.899	2.880 / 4.960 / 9.111
Rotation effective lag	ms	10	252.2 / 260.0 / 270.5	252.2 / 260.0 / 265.0	365.0 / 370.0 / 375.0	345.0 / 372.5 / 393.2
Rotation lag-aligned RMSE	deg	10	2.650 / 3.335 / 3.976	2.650 / 3.281 / 3.981	2.753 / 3.389 / 3.984	3.408 / 4.691 / 5.974
Occlusion episode P95	mm	9	10.963 / 25.553 / 58.004	9.828 / 25.479 / 57.740	4.702 / 11.710 / 26.601	1.659 / 1.980 / 2.985
Start-transition response	ms	9	25.1 / 167.5 / 321.7	159.1 / 208.8 / 355.0	175.9 / 284.4 / 371.6	470.8 / 591.1 / 749.1
Catastrophic occlusion failures (> 40 mm)	count	9	2/9 (max 61.245)	2/9 (max 60.975)	0/9 (max 27.922)	0/9 (max 3.252)